

433-HW9

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Problem 1

```
set.seed(10)
n = 10000
x = runif(n,0,pi/3)
cat('Estimated value:', mean(sin(x))) # true value = 0.5
```

```
## Estimated value: 0.4775508
```

Problem 2

```
set.seed(20)
n = 10000
x = runif(n,0,0.5)
theta_hat = mean(exp(-x))/2
cat('Estimated value:', theta_hat, '\n') # true value = ~0.3934
```

```
## Estimated value: 0.3937203
```

```
cat('Variance of the estimate:', sum((exp(-x)-theta_hat)^2)/n)
```

```
## Variance of the estimate: 0.1680154
```

Problem 3

```
# estimate sqrt(1-x^2) integrated from 0 to 1, true value = ~0.7853
# h_1(x) = 6x(1-x), 0 < x < 1, --> Beta(2,2)
# h_2(x) = 3x^2, 0 < x < 1
g = function(x) {sqrt(1-x^2)}
set.seed(30)
n = 10000
x1 = rbeta(n,2,2) # h_1(x)
mean(g(x1)/dbeta(x1,2,2)) # importance sampling
```

```
## [1] 0.7754069
```

```
u = runif(n)
x2 = u^(1/3) # h_2(x) inverse transform
mean(g(x2)/(3*x2^2))
```

```
## [1] 0.7514618
```

Problem 4

```
g = function(x) {cos(pi*x/2)}
set.seed(40)
n = 10000
x_1 = runif(n) # Monte Carlo
cat('MC estimated value:', mean(cos(pi*x_1/2)), '\n') # true = ~0.6366
```

```
## MC estimated value: 0.6331729
```

```
u = runif(n)
x_2 = u^(1/2) # Importance Sampling
cat('IS estimated value:', mean(g(x_2)/x_2)/2) # inverse transform?
```

```
## IS estimated value: 0.6113074
```